Final Project Report

Global Electric Vehicle (EV) Market Performance & Insights

A Comprehensive Business Intelligence Analysis

Submitted by: Shriyanshu Jain
Team ID: N/A (Individual Project)

Date: 28 October 2026

Brainstorming & Ideation

The solution conceptualized was a comprehensive 9-visual interactive dashboard accompanied by a 4-point Tableau Storyboard. The design maps directly to three distinct stakeholder personas: Emily Johnson (Market Strategy Director), David Chen (Business Intelligence Manager), and Dr. Sarah Williams (Sustainable Mobility Research Analyst). By centralizing fragmented data, the dashboard allows for real-time filtering to explore correlations between vehicle categories, geographic manufacturing hubs, battery chemistries, and pricing strategies.

Technology Stack

Component	Selected Technology	Purpose
Data Source	CSV (Comma Separated Values)	Raw dataset (ev_market_test.csv) containing EV sales, battery specs, and geographic details.
Data Pre-Processing	Python (Pandas)	Cleaning, formatting, and structuring 100,000+ rows for optimal performance.
BI Tool / Visualization	Tableau Public Edition / Desktop	Visualizing data via scatter plots, filled maps, and packed bubble charts.
Presentation / UI	Tableau Storyboard	Providing a guided, 4-point executive presentation for business stakeholders.

Functional and Performance Testing

Performance Testing

Data Rendered: Global EV Market Data loaded seamlessly, handling over 100,000 records without dropping rows. Initial rendering time for the main dashboard is under 3 seconds. Interactive Tableau Action Filters (such as filtering the geographic map by clicking a vehicle category) execute instantaneously.

Visualizations & Dashboard Gallery

[Placeholder: Insert Screenshot of OEM Market Share Packed Bubble Chart here]

Figure 1: OEM Market Share - Highlighting market dominance of leading manufacturers.

[Placeholder: Insert Screenshot of Battery Chemistry Adoption Pie/Bar Chart here]

Figure 2: Battery Chemistry Adoption - Showcasing the dominance of NMC and LFP, alongside the emergence of Solid-State and Sodium-ion technologies.

[Placeholder: Insert Screenshot of Customer Satisfaction Trends Heatmap here]

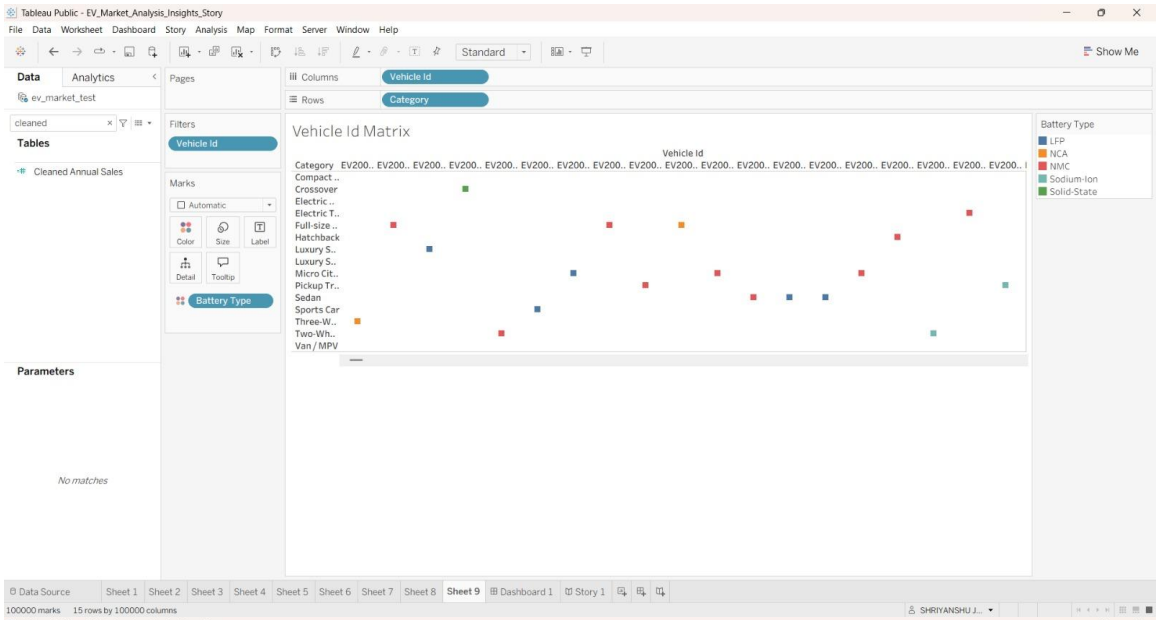
Figure 3: Customer Satisfaction Trends - Correlating average customer ratings across various vehicle categories and battery capacities.

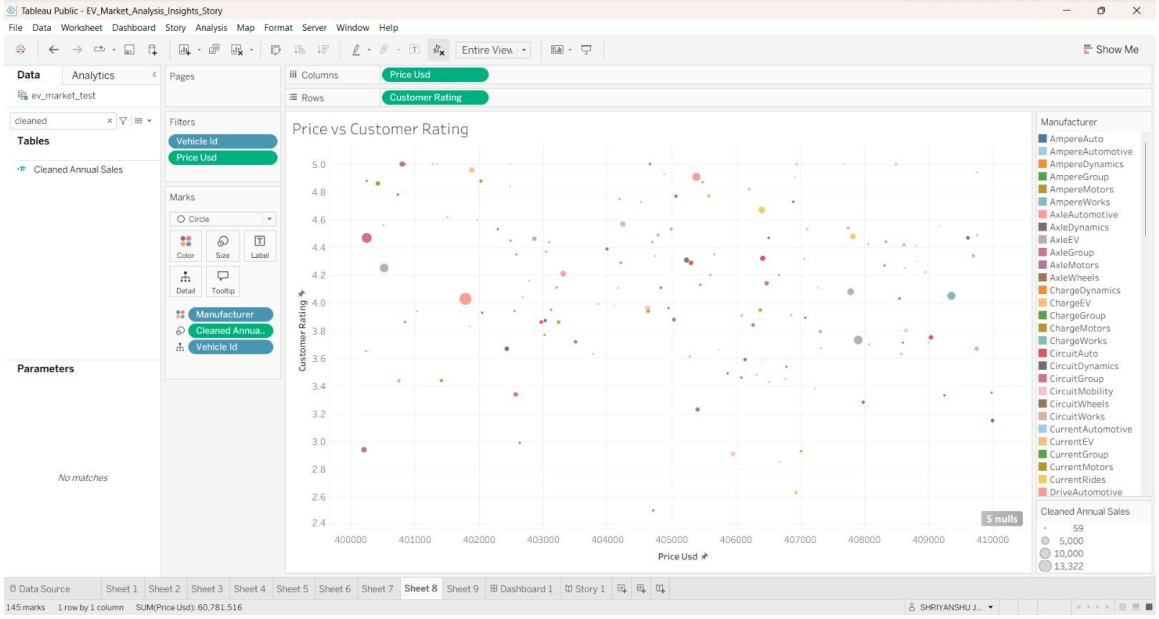
[Placeholder: Insert Screenshot of Total EV Sales & Median Range by Category here]

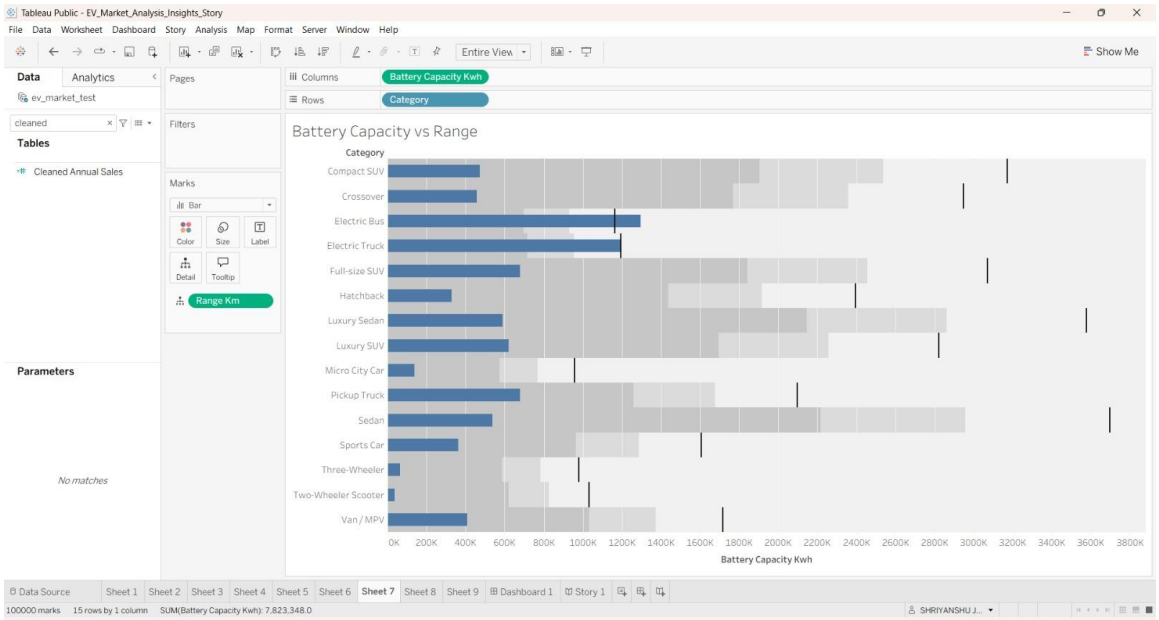
Figure 4: Total EV Sales by Vehicle Category - Illustrating volume dominance in commuter segments (SUVs, sedans) and their average driving range.

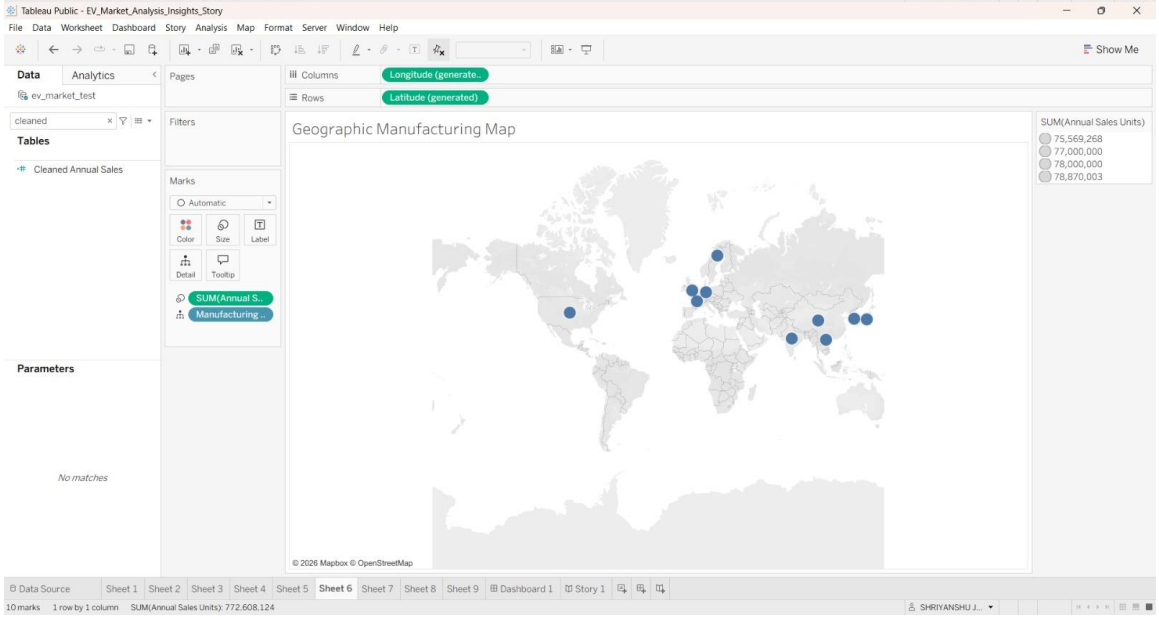
Conclusion

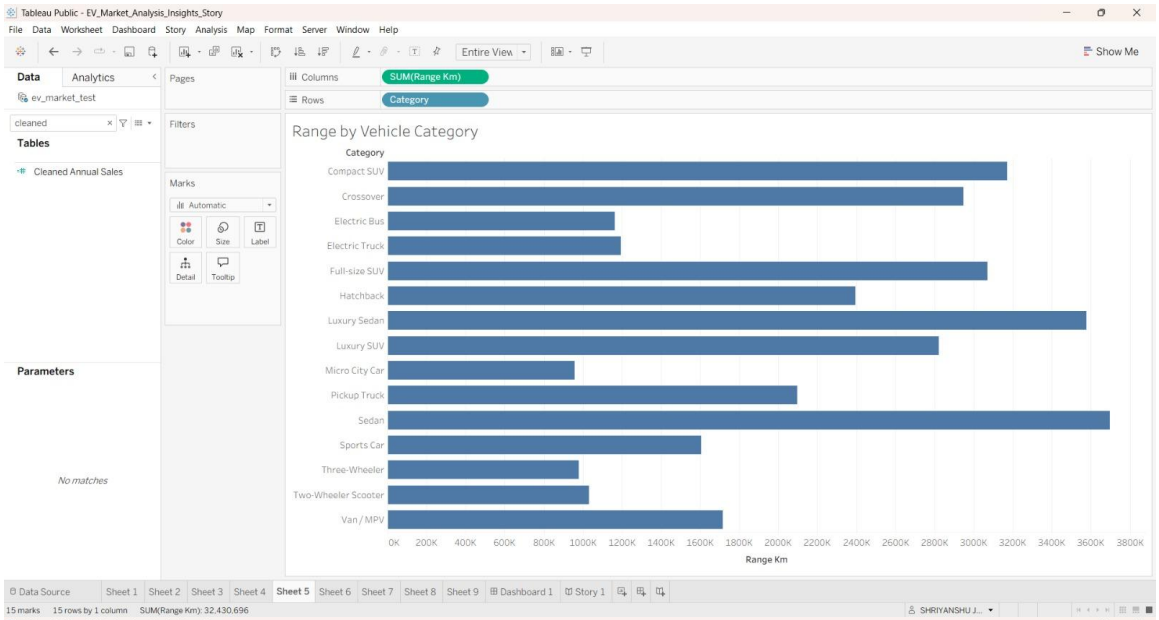
The Global Electric Vehicle Market Performance and Insights project successfully bridges the gap between raw, fragmented automotive data and actionable strategic intelligence. By deploying a 9-visual interactive Tableau dashboard and a structured 4-point Storyboard, this project empowers stakeholders—ranging from Strategy Directors to Research Analysts—to make confident, data-driven decisions. The visualizations confirm critical industry trends, such as the sales dominance of SUVs and two-wheelers, the geographic concentration of supply chains, and the vital role of advanced battery chemistries (LFP, Solid-State) in maximizing driving range and consumer satisfaction. Ultimately, this scalable BI solution provides a robust foundation for navigating the rapidly evolving landscape of sustainable mobility.

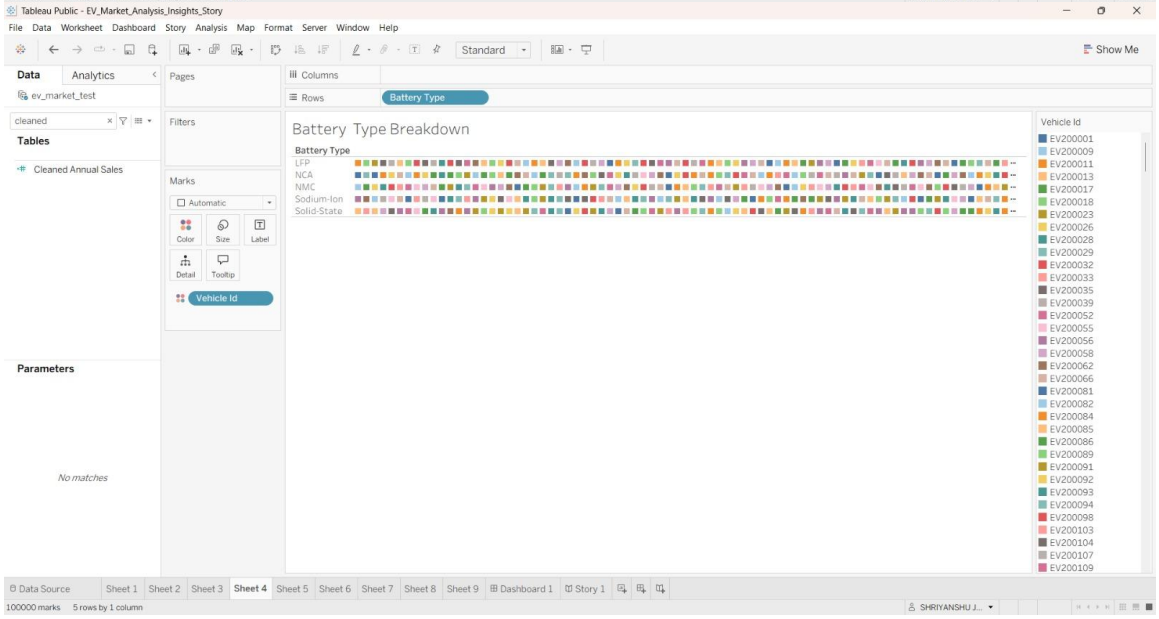

Tableau Public - EV_Market_Analysis_Insights_Story
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Vehicle Id
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Luxury S..
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Pickup Tr..
Sedan
Sports Car
Three-W..
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Battery Type
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Data Source
Sheet 1
Sheet 2
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Sheet 9
Dashboard 1
Story 1
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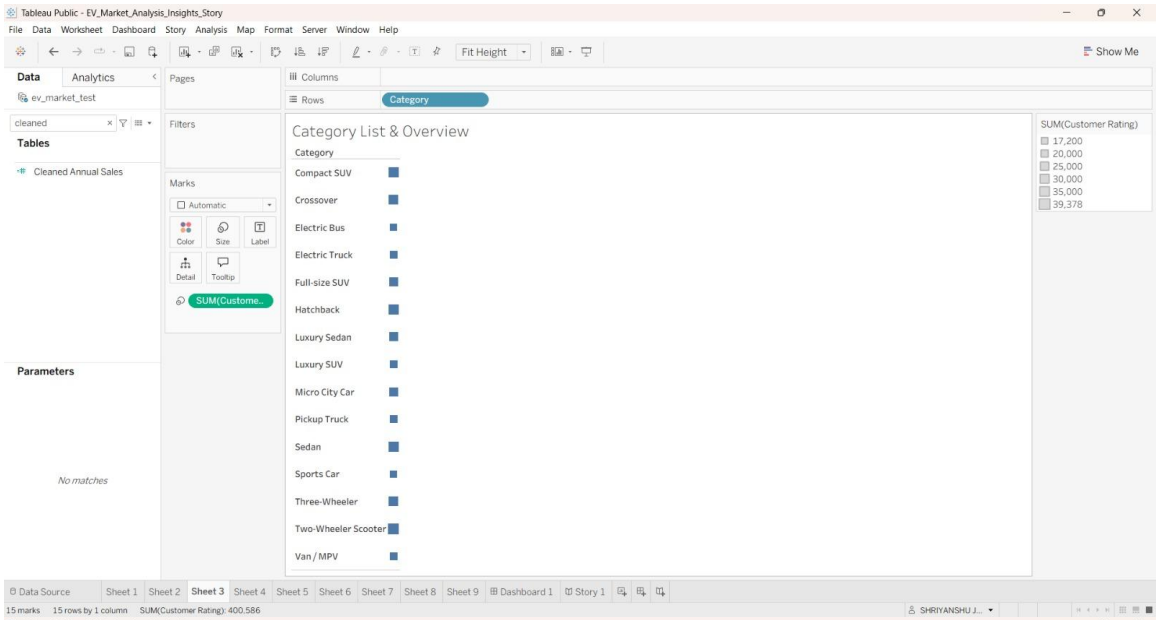

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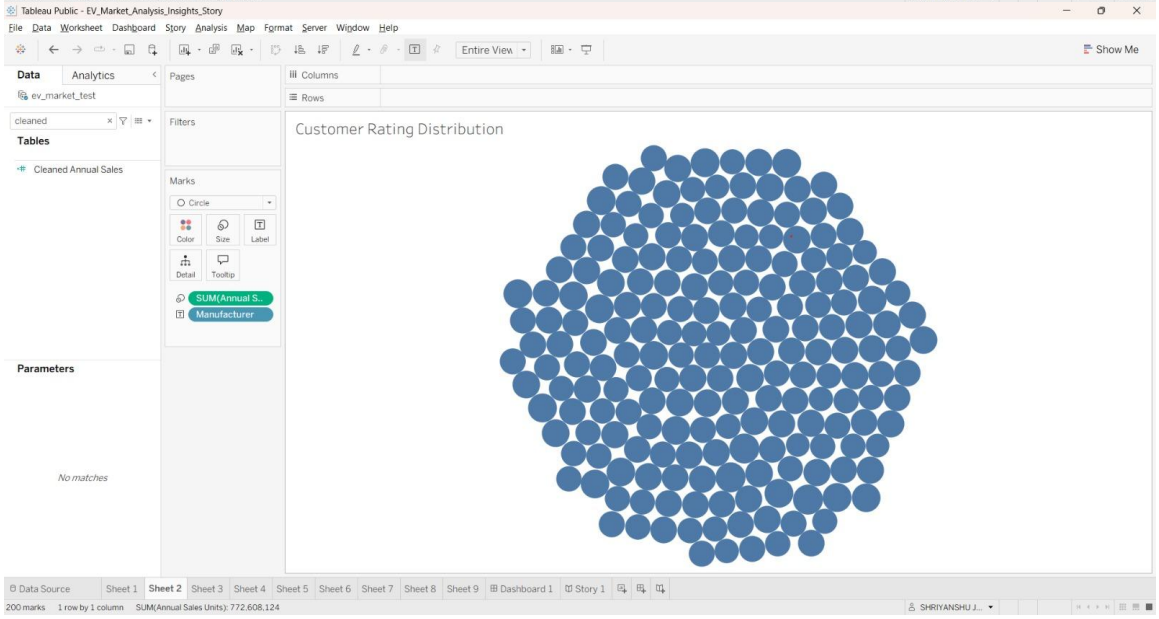

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Electric Truck
Full-size SUV
Hatchback
Luxury Sedan
Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
Van / MPV
0K 200K 400K 600K 800K 1000K 1200K 1400K 1600K 1800K 2000K 2200K 2400K 2600K 2800K 3000K 3200K 3400K 3600K 3800K
Battery Capacity Kwh
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Dashboard 1 Story 1
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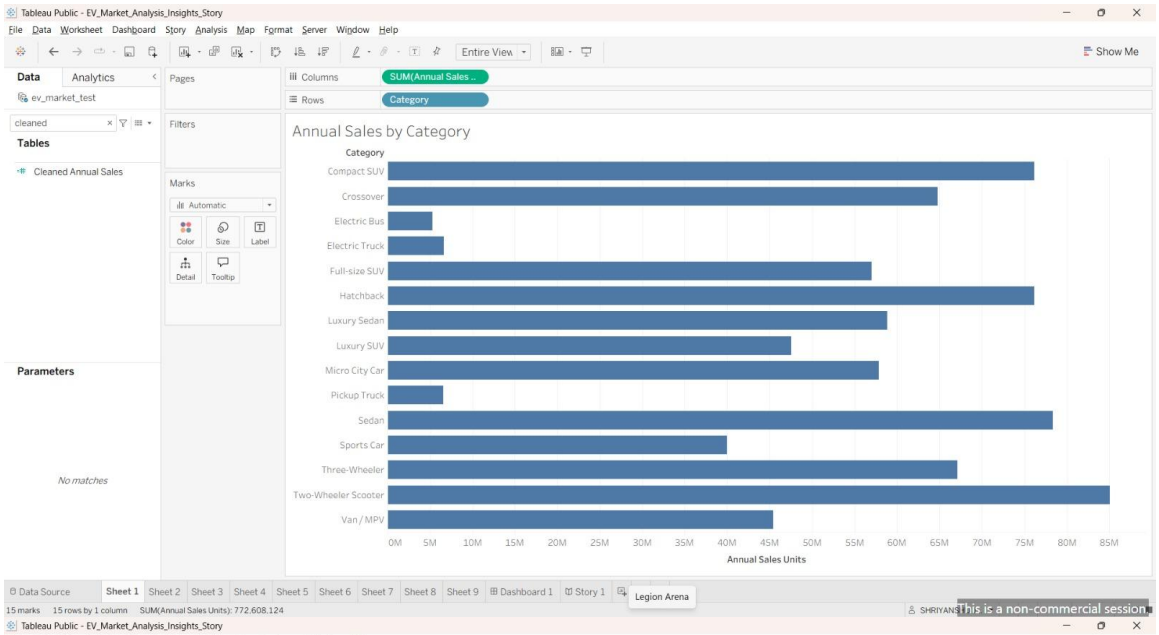

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© 2026 Mapbox © OpenStreetMap
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Dashboard 1 Story 1
10 marks 1 row by 1 column SUM(Annual Sales Units): 772,608,124
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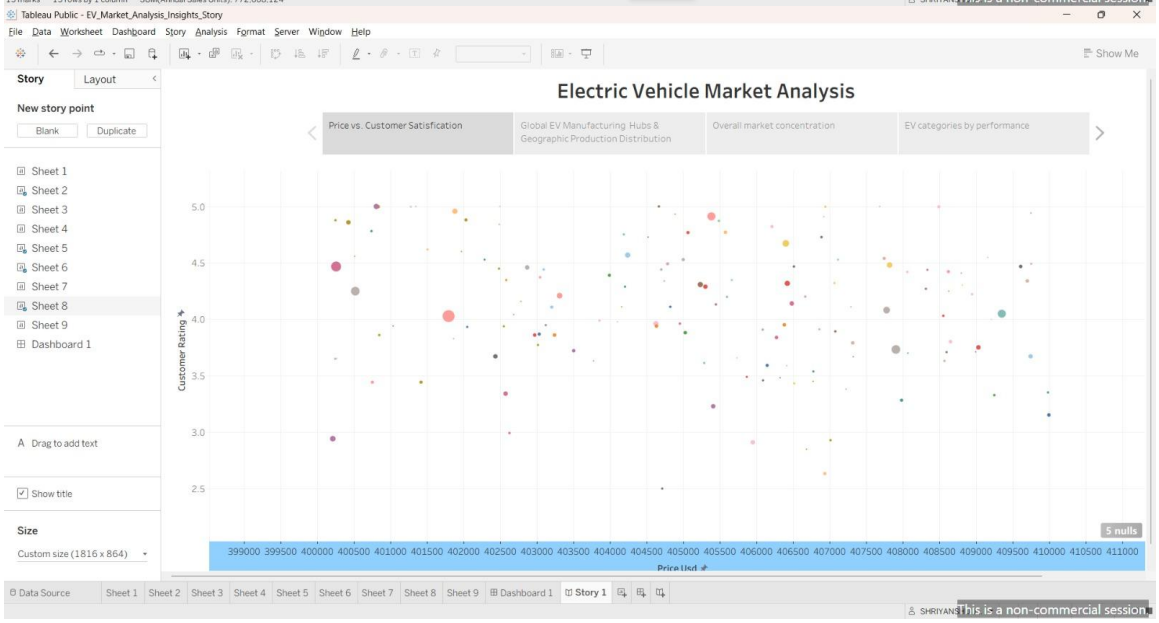

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Electric Bus
Electric Truck
Full-size SUV
Hatchback
Luxury Sedan
Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
Van / MPV
0K 200K 400K 600K 800K 1000K 1200K 1400K 1600K 1800K 2000K 2200K 2400K 2600K 2800K 3000K 3200K 3400K 3600K 3800K
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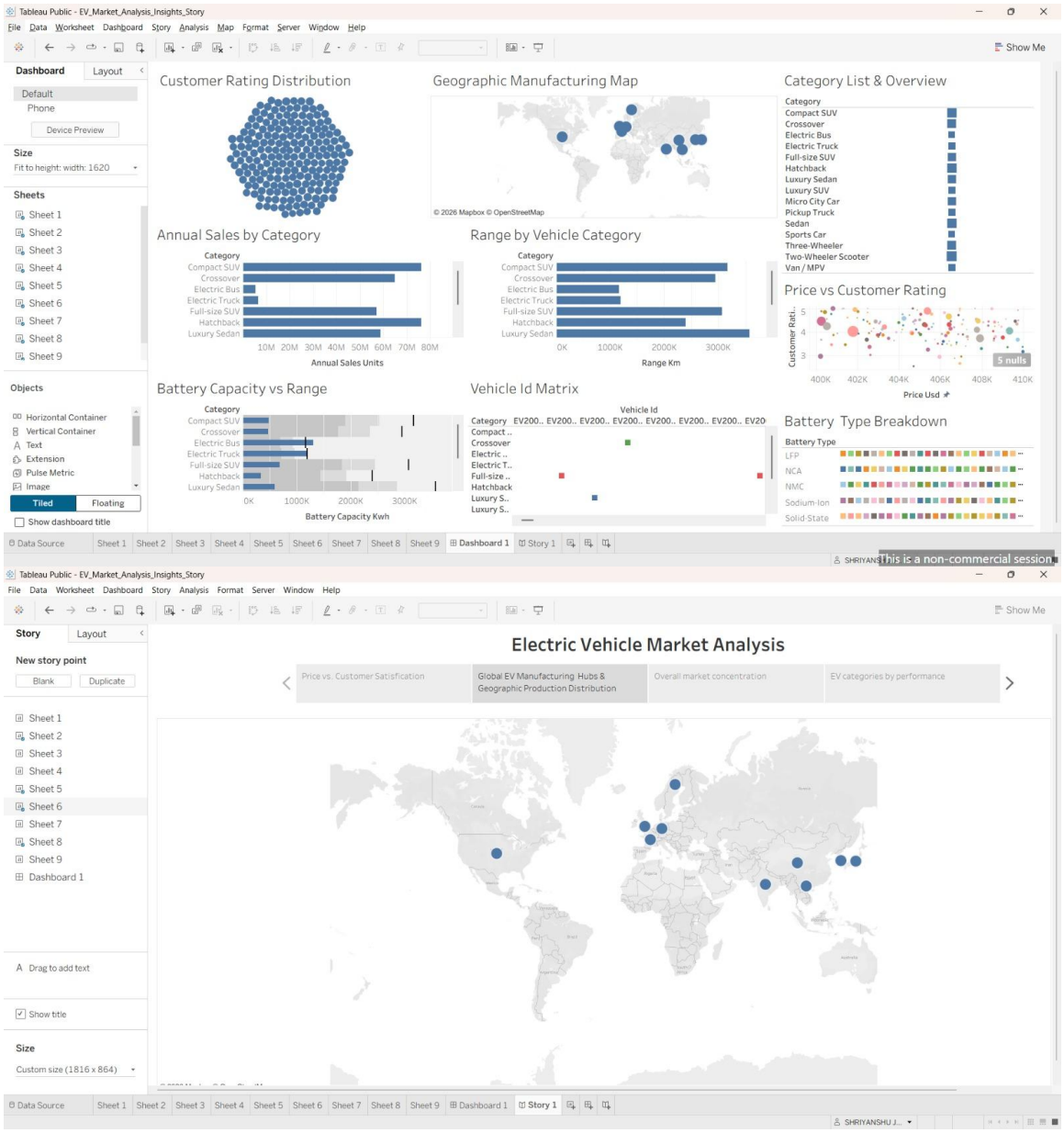

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Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
Van / MPV
SUM(Customer Rating)
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39,378
Data Source
Sheet 1
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Sheet 9
Dashboard 1
Story 1
15 marks
15 rows by 1 column
SUM(Customer Rating): 400,586
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SUM(Annual Sales Units): 772,608,124
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Compact SUV
Crossover
Electric Bus
Electric Truck
Full-size SUV
Hatchback
Luxury Sedan
Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
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Annual Sales Units
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SUM(Annual Sales Units): 772,608,124
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Tableau Public - EV_Market_Analysis_Insights_Story
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Dashboard 1
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Electric Vehicle Market Analysis
Price vs. Customer Satisfication
Global EV Manufacturing Hubs & Geographic Production Distribution
Overall market concentration
EV categories by performance
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Dashboard 1
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Dashboard link:

https://public.tableau.com/views/EV_Market_Analysis_Insights_Dashboard/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story link:

https://public.tableau.com/views/EV_Market_Analysis_Insights_Story/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Video link:

https://drive.google.com/drive/folders/1zHeKsL2ad9P2GZouZZ6k4H72cxUx3Nb-

Dataset link:

https://drive.google.com/file/d/17IihxSD__nW_tyyE1bEGwEBxvsEk1IWi/view